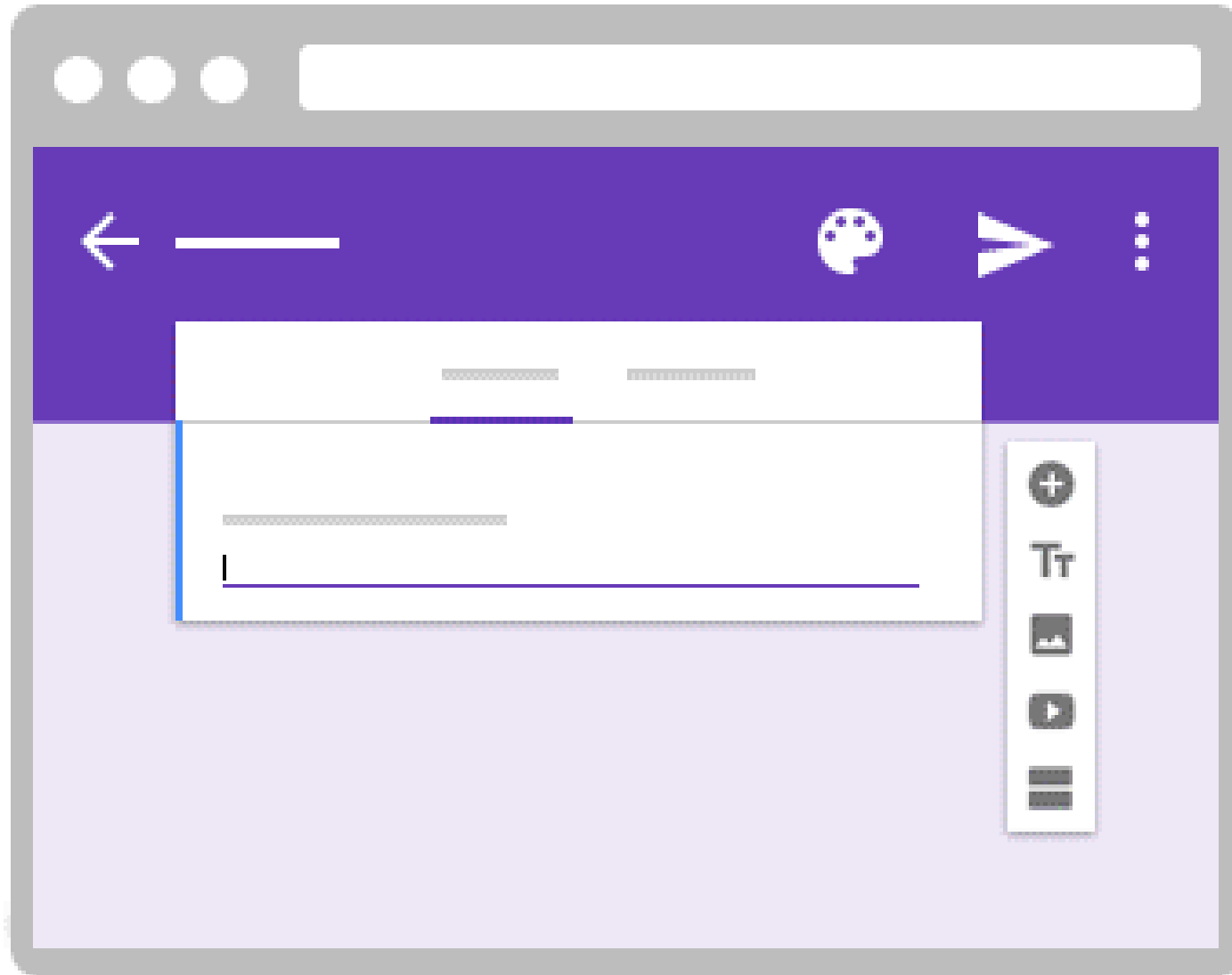


surveydown: **An Open-Source, Markdown-Based Platform for Interactive and Reproducible Surveys**

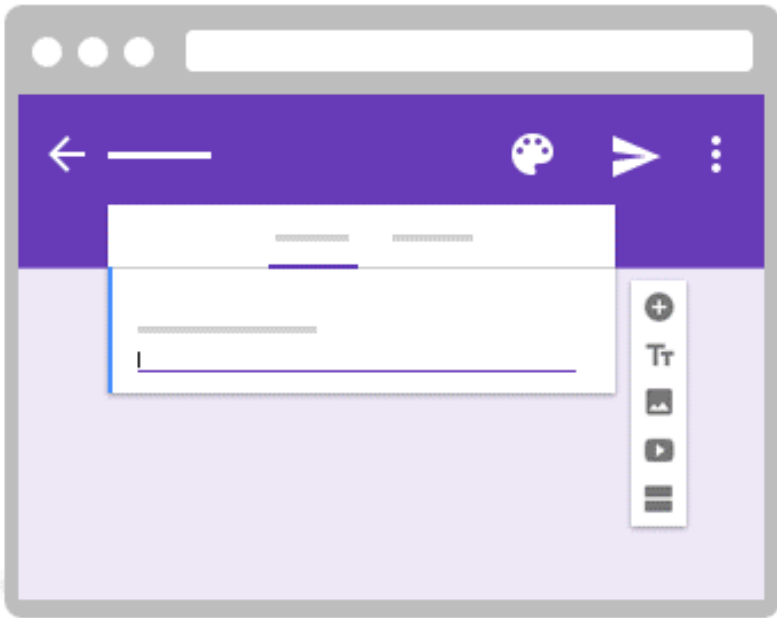
Pingfan Hu, Bogdan Bunea, John Paul Helveston



Typical experience making a survey



Typical experience making a survey



✗ Reproducible

✗ Data Control

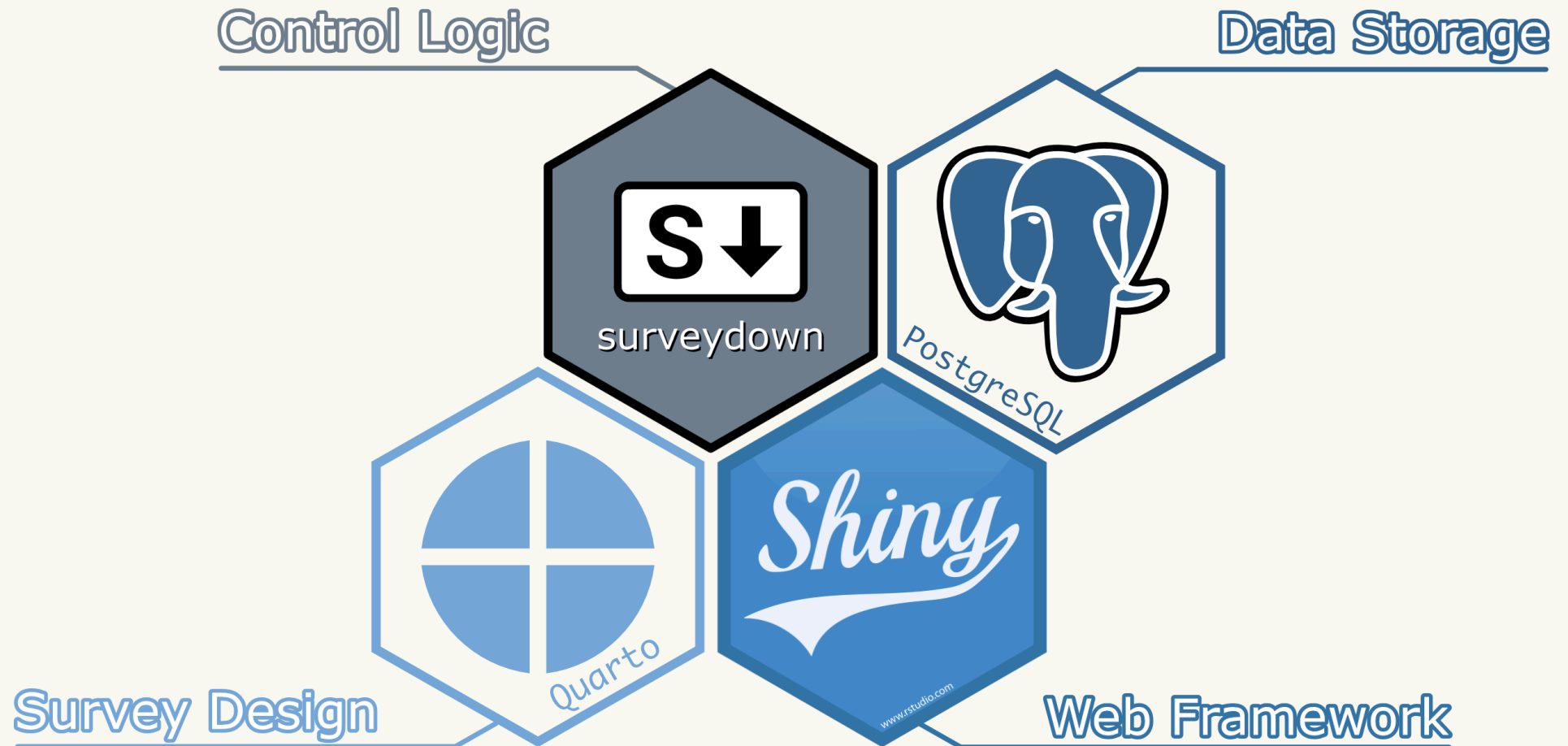
⚠ Free to Use

⚠ Open Source

⚠ Easy Collaboration

⚠ Feature Packed

Introducing surveydown



Components

`survey.qmd`

A **Quarto doc** defining the main survey content (pages, text, images, questions, etc).

`app.R`

An **R script** defining the survey Shiny app. It sets up the database and server, and launches the survey.

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 ::: {#welcome .sd-page}
12
13 # Welcome to our survey!
14
15 ```{r}
16 sd_question(
17   type = "mc",
18   id   = "penguins",
19   label = "What's your favorite penguin?",
20   option = c(
21     "Adélie"   = "adelie",
22     "Chinstrap" = "chinstrap",
23     "Gentoo"   = "gentoo"
24   )
25 )
26
27 sd_next()
28 ```
29
30 :::
31
32 ::: {#end .sd-page}
33
34 This is the last page of the survey.
35
36 ```{r}
37 sd_close()
38 ```
39
```

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
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27 sd_next()
28 ```
29
30 :::
31
32 ::: {#end .sd-page}
33
34 This is the last page of the survey.
35
36 ```{r}
37 sd_close()
38 ```
39
```

YAML Header

Load Package

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
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7 ```{r}
8 library(surveydown)
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27 sd_next()
28 ```
29
30 :::
31
32 ::: {#end .sd-page}
33
34 This is the last page of the survey.
35
36 ```{r}
37 sd_close()
38 ```
39
```

Define pages using “fences”:

```
::: {#id .sd-page}
:::
```

Page 1 (Welcome)

Page 2 (End)

survey.qmd

```
1 ---
2 format: html
3 echo: false
4 warning: false
5 ---
6
7 ```{r}
8 library(surveydown)
9 ```
10
11 ::: {#welcome .sd-page}
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21     "Adélie"   = "adelie",
22     "Chinstrap" = "chinstrap",
23     "Gentoo"   = "gentoo"
24   )
25 )
26
27 sd_next()
28 ```
29
30 :::
31
32 ::: {#end .sd-page}
33
34 This is the last page of the survey.
35
36 ```{r}
37 sd_close()
38 ```
39
```

Contents of Page 1

survey.qmd

```
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5 ---
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27 sd_next()
28 ```
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30 :::
31
32 ::: {#end .sd-page}
33
34 This is the last page of the survey.
35
36 ```{r}
37 sd_close()
38 ```
39
```

Question Definition

app.R

```
1 # Load Package
2 library(surveydown)
3
4 # Database Credentials
5 sd_db_config()
6
7 # Connect to Database
8 db <- sd_db_connect()
9
10 # Server Setup
11 server <- function(input, output, session) {
12
13   # Skip logic
14   sd_skip_forward(...)
15
16   # Conditional display
17   sd_show_if(...)
18
19   # Server settings
20   sd_server(
21     db = db,
22     ...
23   )
24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
```

app.R

```
1 # Load Package
2 library(surveydown)
3
4 # Database Credentials
5 sd_db_config()
6
7 # Connect to Database
8 db <- sd_db_connect()
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11 server <- function(input, output, session) {
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14   sd_skip_forward(...)
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18
19   # Server settings
20   sd_server(
21     db = db,
22     ...
23   )
24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
```

Load Package

```
1 # Load Package
2 library(surveydown)
3
4 # Database Credentials
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20   sd_server(
21     db = db,
22     ...
23   )
24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
```

Database Setup

We recommend using **Supabase** or any **PostgreSQL** database for the database setup.

app.R

```
1 # Load Package
2 library(surveydown)
3
4 # Database Credentials
5 sd_db_config()
6
7 # Connect to Database
8 db <- sd_db_connect()
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21     db = db,
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24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
```

Server Setup

app.R

```
1 # Load Package
2 library(surveydown)
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19   # Server settings
20   sd_server(
21     db = db,
22     ...
23   )
24 }
25
26 # Launch Survey
27 shiny::shinyApp(ui = sd_ui(), server = server)
```

Launch Survey

Locally running the survey

The screenshot displays the RStudio environment with a script titled 'survey.qmd' open in the Source editor. The script is a Quarto document that sets up a survey using the 'surveydown' package. It includes a title 'demo-survey', a format of 'html', and a question about favorite penguins. The console on the right shows the output of the script, which includes a welcome message and a list of penguin species. The Files pane on the right shows the project structure, including 'app.R', 'survey.qmd', and 'survey.Rproj'.

```
1 ---  
2 format: html  
3 echo: false  
4 warning: false  
5 ---  
6  
7 ```{r}  
8 library(surveydown)  
9 ```  
10  
11 ::: {#welcome .sd-page}  
12  
13 # Welcome to our survey!  
14  
15 ```{r}  
16 sd_question(  
17   type = 'mc',  
18   id = 'penguins',  
19   label = "What's your favorite penguin?",  
20   option = c(  
21     'Adélie' = 'adelie',  
22     'Chinstrap' = 'chinstrap',  
23     'Gentoo' = 'gentoo'  
24   )  
25 )  
26  
27 sd_next()  
28 ```  
29  
30 :::  
31  
32 ::: {#end .sd-page}  
33  
34 The next page of the survey.  
35
```

Console output:

```
R 4.3.0 - ~/Documents/Codes/demo-survey/  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
> |
```

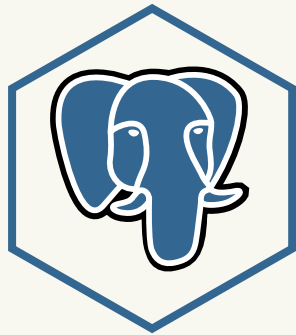
Name	Size	Modified
..		
app.R	399 B	Jan 12, 2025, 10:44 AM
survey.qmd	446 B	Jan 12, 2025, 10:42 AM
survey.Rproj	205 B	Jan 12, 2025, 12:54 PM

Data saved locally in csv for previewing purposes.

Online Deployment

Data Storage

- Any PostgreSQL database
- Recommend **Supabase** (Free)
- Use `sd_db_config()` to set up



PostgreSQL



Supabase

Deploy

- Recommend **shinyapps.io** (Free):

```
rsconnect::deployApp(appName = "my_survey")
```

- Other Platforms:
 - Posit Connect Cloud
 - Hugging Face
 - Heroku

Online Data Storage + Online Survey URL

Local Dashboard App

The screenshot shows the RStudio interface for a project named 'question-types'. The editor window displays the following R script:

```
1 # Package setup -----
2
3 # Install required packages:
4 # install.packages("pak")
5 # pak::pak('surveydown-dev/surveydown') # Development version from github
6
7 # Load packages
8 library(surveydown)
9
10 # Database setup -----
11 #
12 # Details at: https://surveydown.org/manuals/storing-data
13 #
14 # surveydown stores data on any PostgreSQL database. We recommend
15 # https://supabase.com/ for a free and easy to use service.
16 #
17 # Once you have your database ready, run the following function to store your
18 # database configuration parameters in a local .env file:
19 #
20 # sd_db_config()
21 #
22 # Once your parameters are stored, you are ready to connect to your database.
23 # For this demo, we set ignore = TRUE in the following code, which will ignore
24 # the connection settings and won't attempt to connect to the database. This is
25 # helpful if you don't want to record testing data in the database table while
26 # doing local testing. Once you're ready to collect survey responses, set
27 # ignore = FALSE or just delete this argument.
28
29 db <- sd_db_connect()
30
31 # Server setup -----
32
33 0:00 / 0:59
34 8:1 Package setup
```

The console window is empty. The file explorer shows the project structure:

Name	Size	Modified
..		
app.R	1.7 KB	Mar 20, 2025, 10:08 PM
survey.Rproj	414 B	Mar 21, 2025, 4:21 PM
preview_data.csv	1.8 KB	Mar 20, 2025, 1:39 PM
.gitignore	5 B	Mar 19, 2025, 9:26 AM
survey.qmd	9.6 KB	Jan 22, 2025, 8:34 AM
_survey		
rsconnect		

Templates and demos are provided on
surveydown.org for an easy start.

Summary of Features

Reactivity and randomization

Dashboard for survey data

Cookies for restoration

URL parameters for redirection

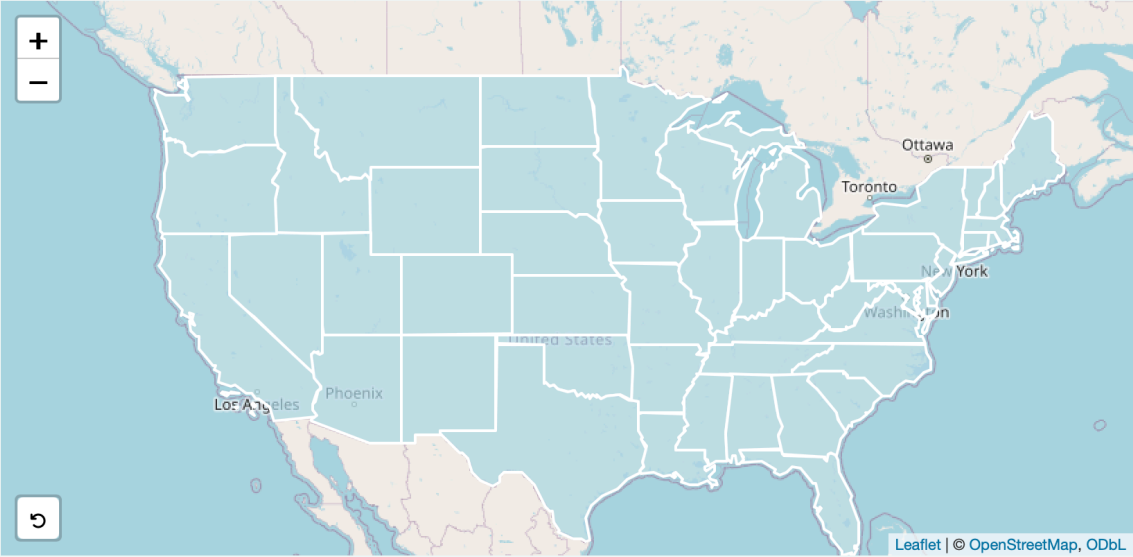
Ability to start from selected page

Custom interactive questions

Example of custom interactive question

Demo - Custom Leaflet Map

Click on the state you live in:



You live in:

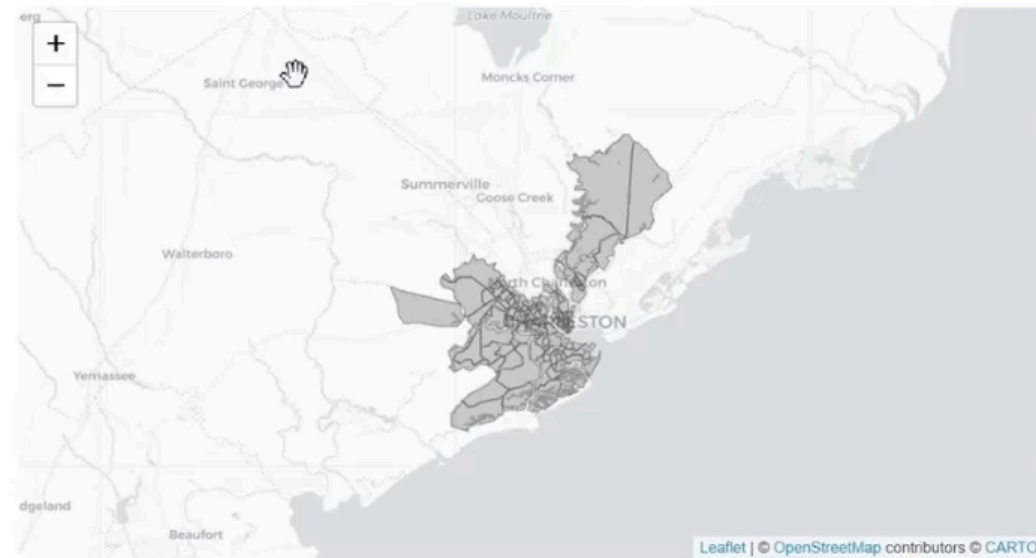
Next

Made with [surveydown](#) [Source Code](#)

Map Question Showcase

Question 1: Social Vulnerability is how likely people are to face greater harm or impact on their lives after disruptive hazard events like flooding based on social or economic conditions like income or transportation access.

Please review the map of the City of Charleston, South Carolina, and select the areas you believe are more socially vulnerable to the impacts of hazardous events.



Next Question

▶ 0:00 / 0:56



Source: Dr. Selena Hinojos, GWU

Future Work

GUI for defining surveys

More question types

...and more!

You can contribute!

Check out surveydown.org for more!